# IGCSE 0607 Practice Questions — E6.5

## Compound shapes and parts of shapes

**Total Questions:** 30 — **Total Marks:** 219 Marks

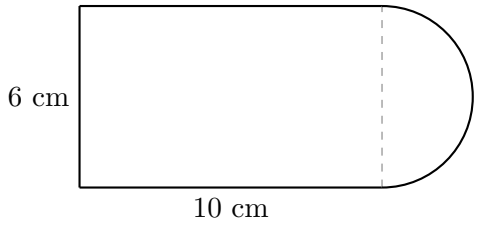
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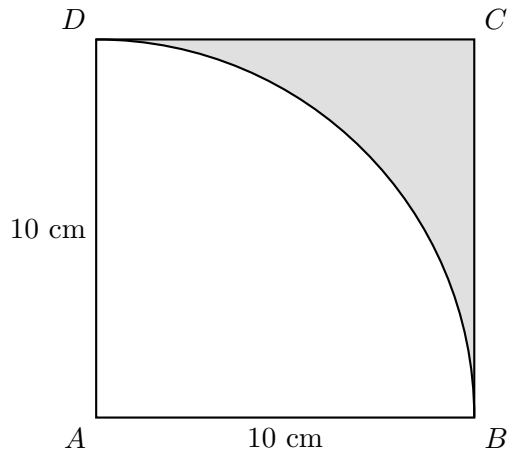
**1** A shape is made from a rectangle and a semicircle.
The rectangle has a length of 10 cm and a width of 6 cm.
The semicircle is attached to one of the 6 cm sides.




Find the exact perimeter of the shape.
Give your answer in the form $a + b\pi$.

..... cm **[3]**

**2** The diagram shows a square $ABCD$ with side length 10 cm.
A quarter circle with center $A$ and radius 10 cm is drawn inside the square.
The region inside the square but outside the quarter circle is shaded.



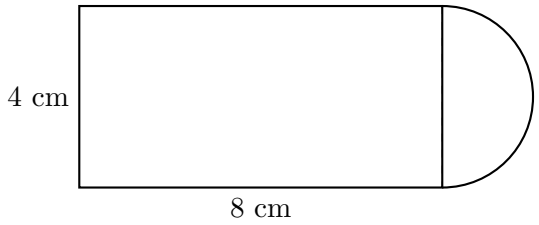

**(a)** Find the exact perimeter of the shaded region.
Give your answer in terms of $\pi$.

..... cm **[2]**

**(b)** Find the exact area of the shaded region.
Give your answer in terms of $\pi$.

..... $cm^2$ **[2]**

**3** The diagram shows a compound shape formed by joining a semi-circle to a rectangle.
The rectangle has a length of 8 cm and a width of 4 cm.
The diameter of the semi-circle is 4 cm.



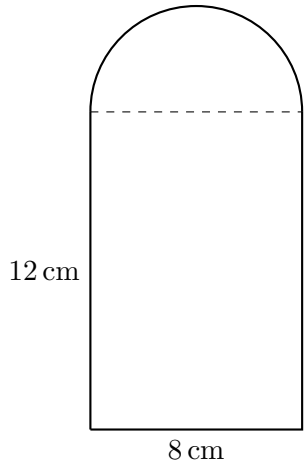

**(a)** Calculate the perimeter of the shape.
Give your answer in its simplest form, in terms of $\pi$.

..... *cm* **[2]**

**(b)** Calculate the area of the shape.
Give your answer in its simplest form, in terms of $\pi$.

..... $cm^2$ **[2]**

**4** A compound shape is made from a rectangle and a semicircle.
The rectangle has width 8 cm and length 12 cm. The semicircle is attached to one of the sides of width 8 cm.



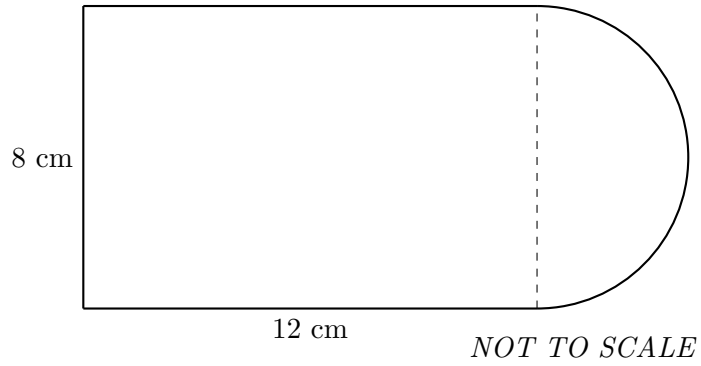

**(a)** Calculate the exact perimeter of the compound shape.
Give your answer in its simplest form, in terms of $\pi$.

..... cm **[2]**

**(b)** Calculate the exact area of the compound shape.
Give your answer in its simplest form, in terms of $\pi$.

..... $cm^2$ **[2]**

**5** The diagram shows a shape made from a rectangle and a semicircle.
The rectangle has length 12 cm and width 8 cm.
The semicircle has a diameter of 8 cm.



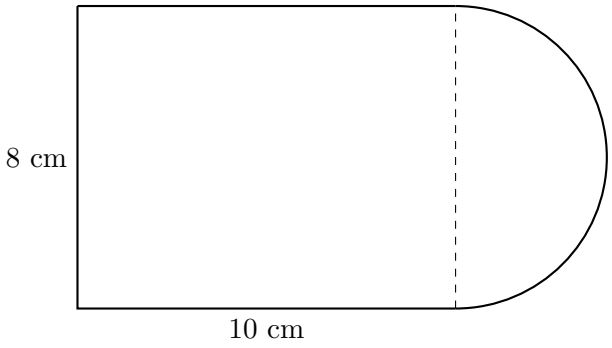

**(a)** Find the perimeter of the shape.
Give your answer in its simplest form in terms of $\pi$.

Perimeter = ..... cm **[2]**

**(b)** Find the area of the shape.
Give your answer in its simplest form in terms of $\pi$.

Area = ..... $cm^2$ **[3]**

**6** The diagram shows a compound shape made from a rectangle and a semicircle.
The rectangle has a length of 10 cm and a width of 8 cm.
The semicircle is attached to one of the 8 cm sides.



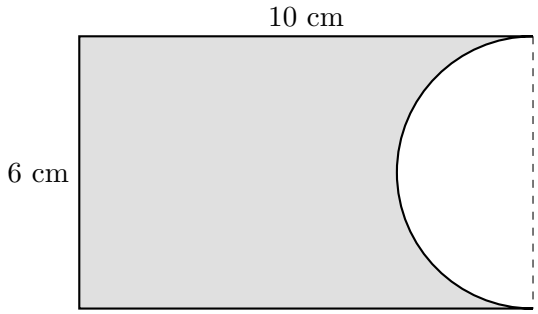

**(a)** Work out the exact area of the shape.
Give your answer in the form $a + b\pi$, where $a$ and $b$ are integers.

Area = ..... $cm^2$ **[3]**

**(b)** Work out the exact perimeter of the shape.
Give your answer in the form $c + d\pi$, where $c$ and $d$ are integers.

Perimeter = ..... cm **[3]**

**7** The diagram shows a compound shape made from a rectangle and a semicircle.
The rectangle has a length of 10 cm and a width of 6 cm.
A semicircle of diameter 6 cm has been removed from one end of the rectangle.




NOT TO SCALE

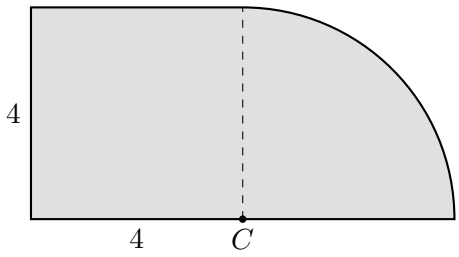
**(a)** Work out the perimeter of the shape.
Give your answer in the form $a + b\pi$, where $a$ and $b$ are integers.

..... cm **[3]**

**(b)** Work out the area of the shape.
Give your answer in the form $c - d\pi$, where $c$ is an integer and $d$ is a decimal.

..... $cm^2$ **[3]**

**8** The diagram shows a shape formed by a square of side 4 cm and a quarter of a circle with center $C$.



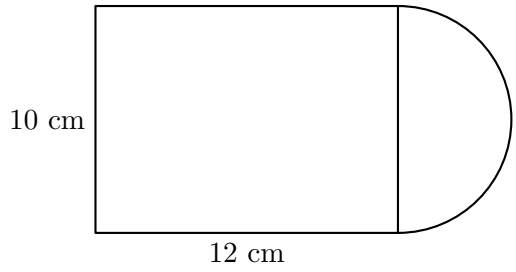

**(a)** Find the area of the shape.
Give your answer in terms of $\pi$.

Area = ..... $cm^2$ **[3]**

**(b)** Find the perimeter of the shape.
Give your answer in terms of $\pi$.

Perimeter = ..... $cm$ **[3]**

**9** The diagram shows a compound shape made from a rectangle and a semicircle.
The rectangle has length 12 cm and width 10 cm.
The semicircle has a diameter of 10 cm.



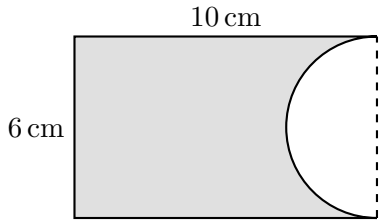

**(a)** Calculate the perimeter of the shape.
Give your answer in the form $a + b\pi$.

..... cm **[3]**

**(b)** Calculate the area of the shape.
Give your answer in the form $c + d\pi$.

..... $cm^2$ **[3]**

**10** The diagram shows a shape made from a rectangle with dimensions 10 cm by 6 cm. A semicircle of diameter 6 cm has been removed from one end.




NOT TO SCALE

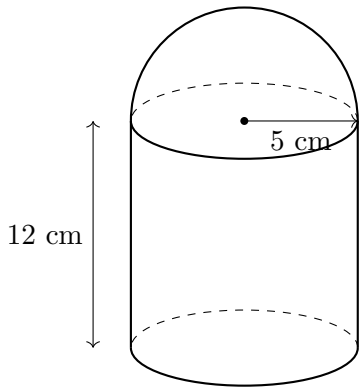
**(a)** Calculate the area of the shape.
Give your answer in terms of $\pi$.

..... $cm^2$ **[3]**

**(b)** Calculate the perimeter of the shape.
Give your answer in terms of $\pi$.

..... cm **[3]**

**11** A solid is made by attaching a hemisphere of radius 5 cm to one end of a solid cylinder.
The cylinder has a radius of 5 cm and a height of 12 cm.



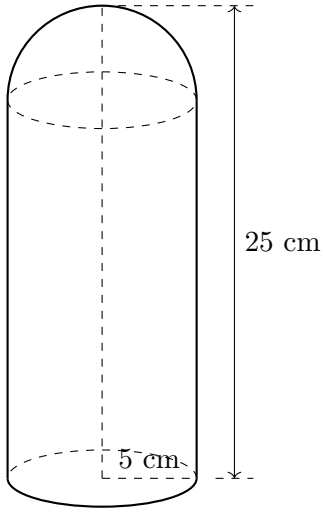

**(a)** Calculate the total volume of the solid.
Give your answer correct to 3 significant figures.

..... $cm^3$ **[3]**

**(b)** Calculate the total surface area of the solid.
Give your answer correct to 3 significant figures.

..... $cm^2$ **[3]**

**12** The diagram shows a solid in the shape of a cylinder with a hemisphere attached to the top.
The radius of both the cylinder and the hemisphere is 5 cm.
The total height of the solid is 25 cm.




NOT TO SCALE

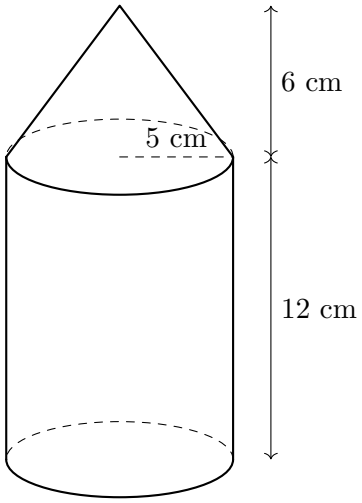
**(a)** Calculate the total volume of the solid.

..... $cm^3$ **[3]**

**(b)** Calculate the total surface area of the solid.

..... $cm^2$ **[4]**

**13** The diagram shows a solid compound shape made from a cylinder and a cone.
The cylinder and the cone both have a base radius of 5 cm.
The height of the cylinder is 12 cm and the height of the cone is 6 cm.



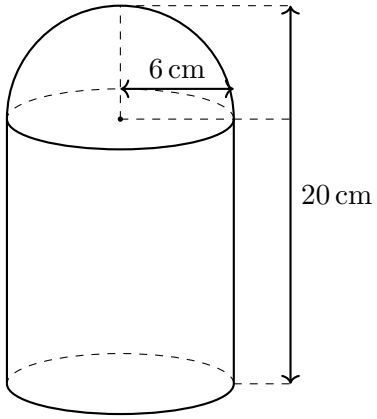

**(a)** Calculate the volume of the solid.

..... $cm^3$ **[3]**

**(b)** Calculate the total surface area of the solid.

..... $cm^2$ **[4]**

**14** A solid is made by placing a solid hemisphere on top of a solid cylinder.
The cylinder and the hemisphere both have radius 6 cm.
The total height of the solid is 20 cm.



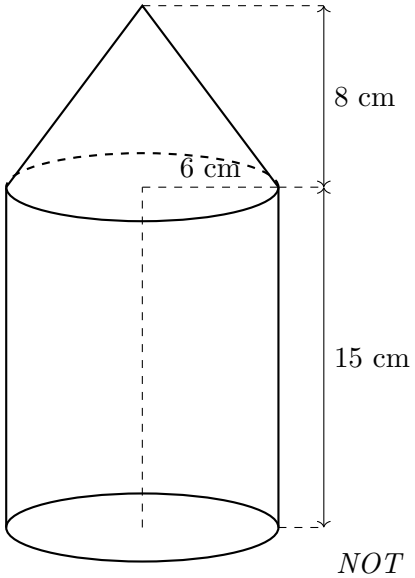

**(a)** Calculate the volume of the solid.

..... $cm^3$ **[3]**

**(b)** Calculate the total surface area of the solid.

..... $cm^2$ **[4]**

**15** A solid is made from a cylinder and a cone.
The cylinder has radius 6 cm and height 15 cm.
The cone has radius 6 cm and height 8 cm.




*NOT TO SCALE*

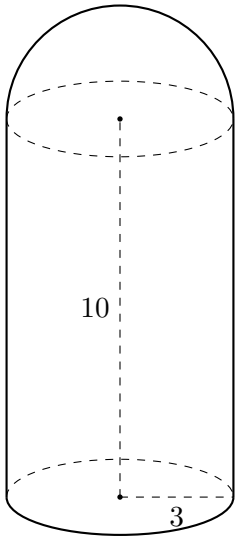
**(a)** Calculate the total volume of the solid.

Volume = ..... $cm^3$ **[3]**

**(b)** Calculate the total surface area of the solid.

Surface area = ..... $cm^2$ **[4]**

**16** A solid is made by attaching a hemisphere of radius 3 cm to one end of a solid cylinder of radius 3 cm and height 10 cm.



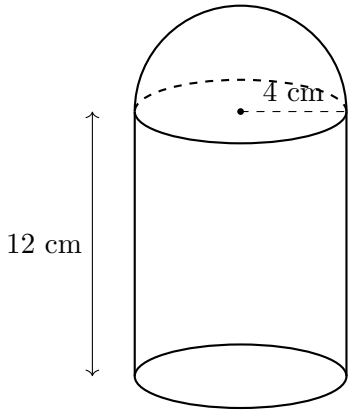

**(a)** Calculate the volume of the solid.

Volume = ..... $cm^3$ **[3]**

**(b)** Calculate the total surface area of the solid.

Total surface area = ..... $cm^2$ **[4]**

**17** A solid object is formed by joining a hemisphere to one end of a solid cylinder.
The cylinder has a radius of 4 cm and a length of 12 cm.
The hemisphere has the same radius as the cylinder.



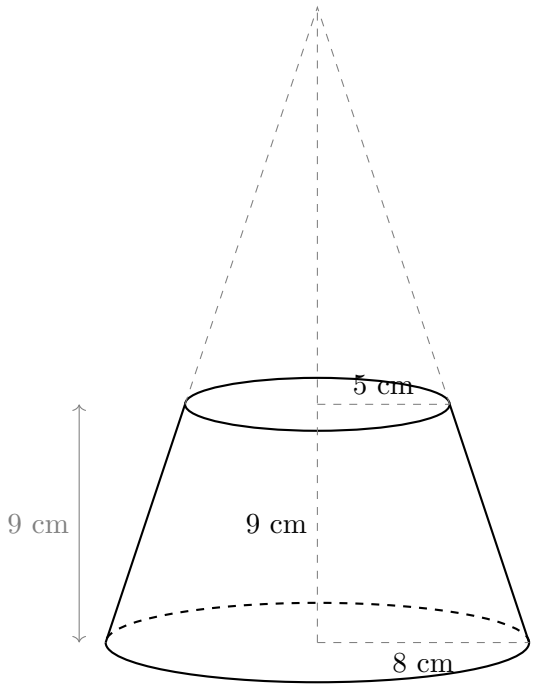

**(a)** Calculate the volume of the solid.

Volume = ..... $cm^3$ **[3]**

**(b)** Calculate the total surface area of the solid.

Surface area = ..... $cm^2$ **[4]**

**18** The diagram shows a solid frustum.
The frustum is made by removing a smaller cone from a larger cone.
The larger cone has a base radius of 8 cm and a vertical height of 24 cm.
The smaller cone has a base radius of 5 cm and a vertical height of 15 cm.



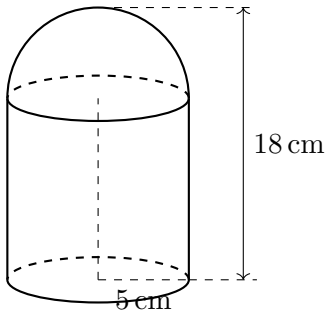

**(a)** Calculate the volume of the solid frustum.

..... $cm^3$ **[3]**

**(b)** Calculate the total surface area of the solid frustum.

..... $cm^2$ **[5]**

**19** A solid is formed by placing a hemisphere of radius 5 cm on top of a cylinder of radius 5 cm.
The total height of the solid is 18 cm.




NOT TO SCALE

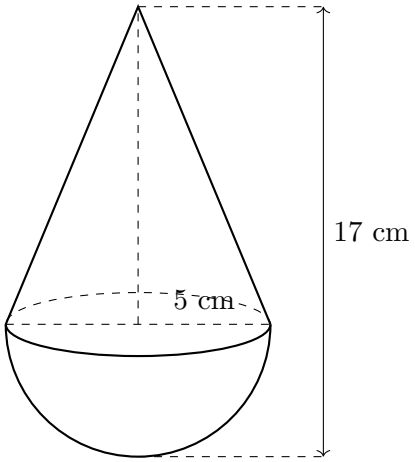
**(a)** Calculate the volume of the solid.
Give your answer correct to 3 significant figures.

..... $cm^3$ **[4]**

**(b)** Calculate the total surface area of the solid.
Give your answer correct to 3 significant figures.

..... $cm^2$ **[4]**

**20** The diagram shows a solid made by joining a solid hemisphere to a solid cone.
The common radius of the hemisphere and the base of the cone is 5 cm.
The total height of the solid is 17 cm.




**(a)** Calculate the volume of the solid.

Volume = ..... $cm^3$ **[3]**

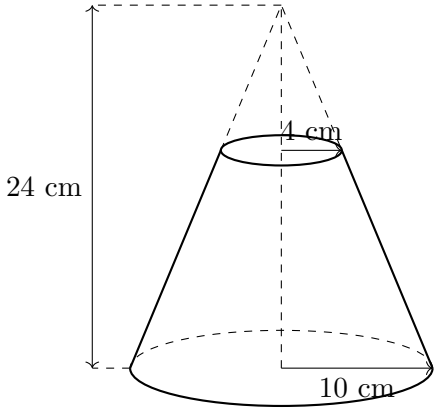
**(b)** Calculate the total surface area of the solid.

Surface area = ..... $cm^2$ **[3]**

**(c)** The solid is melted down and completely recast into a single solid cylinder of radius 4 cm.
Calculate the height of the cylinder.

Height = ..... cm **[3]**

**21** A solid cone of height 24 cm and base radius 10 cm is cut parallel to its base to form a frustum and a smaller cone.
The radius of the top circular face of the frustum is 4 cm.




**(a)** Show that the height of the removed small cone is 9.6 cm.

**[2]**

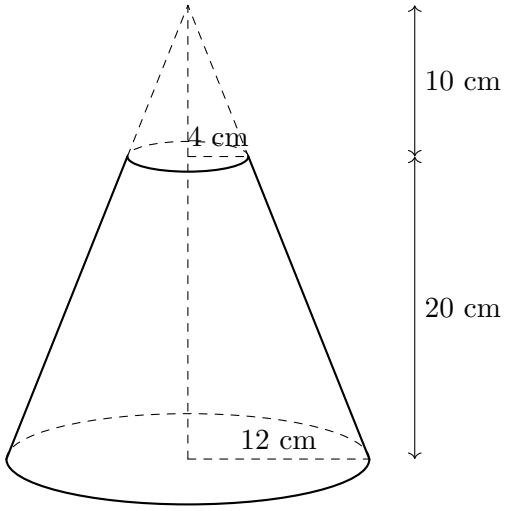
**(b)** Calculate the volume of the frustum.

..... $cm^3$ **[3]**

**(c)** The solid frustum is melted down and recast entirely into a solid sphere.
Calculate the radius of this sphere.

..... $cm$ **[3]**

**22** The diagram shows a solid metal frustum.
The frustum is formed by removing a small cone of height 10 cm from a larger cone.
The radius of the circular base of the frustum is 12 cm and the radius of the circular top is 4 cm.
The height of the frustum is 20 cm.



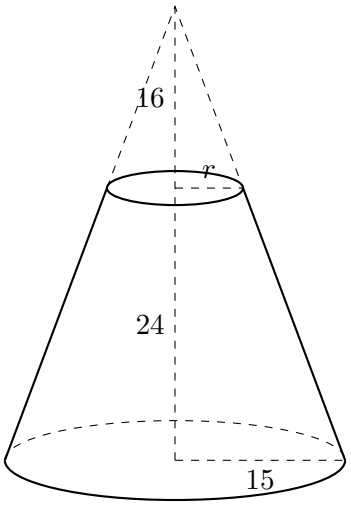

**(a)** Calculate the volume of the solid frustum.

..... $cm^3$ **[4]**

**(b)** Calculate the total surface area of the solid frustum.

..... $cm^2$ **[4]**

**23** A container is in the shape of a frustum of a cone.
The original cone has a base radius of 15 cm and a vertical height of 40 cm.
A smaller cone of vertical height 16 cm is removed from the top to form the frustum.




**(a)** Show that the radius $r$ of the top circular face of the frustum is 6 cm.

**[2]**

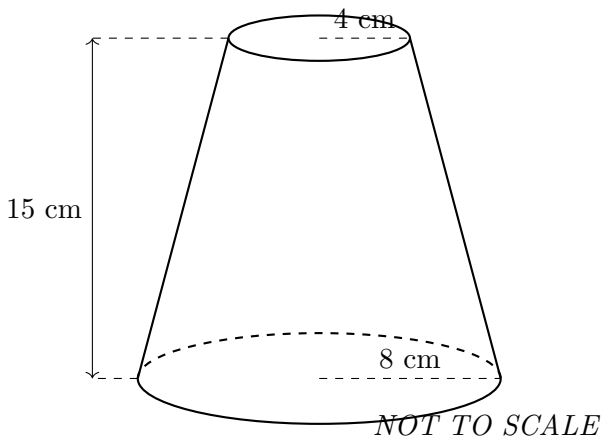
**(b)** Calculate the volume of the frustum.

Volume = ..... $cm^3$ **[3]**

**(c)** The solid frustum is melted down and recast entirely into a solid sphere.
Calculate the radius of the sphere.

Radius = ..... $cm$ **[3]**

**24** A solid frustum of a cone has top radius 4 cm, bottom radius 8 cm and vertical height 15 cm.
It is formed by removing a smaller cone from the top of a larger cone.




**(a)** Show that the original height of the cone before the top was removed was 30 cm.

**[2]**

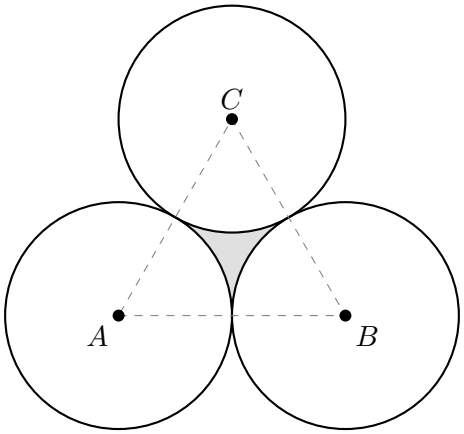
**(b)** Calculate the volume of the frustum.

Volume = ..... $cm^3$ **[3]**

**(c)** Calculate the total surface area of the solid frustum.

Surface area = ..... $cm^2$ **[4]**

**25** The diagram shows three identical circles, each of radius $r$ cm.
The circles touch each other externally.
The region enclosed between the three circles is shaded.




**(a)** Show that the area of the shaded region, $A$, is given by the formula

$$A = r^2\left(\sqrt{3} - \frac{\pi}{2}\right)$$

**[4]**

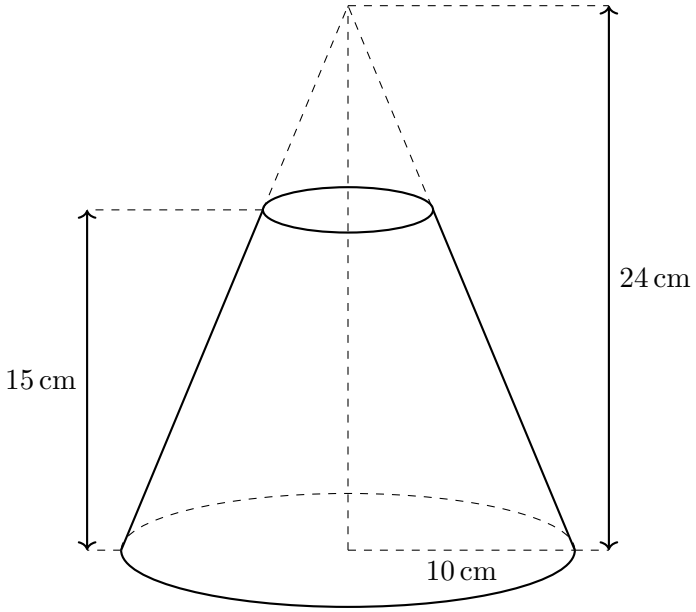
**(b)** The area of the shaded region is 15 cm$^2$.
Work out the value of $r$.

$r =$ ..... **[3]**

**(c)** Calculate the total perimeter of the shaded region.

..... cm **[3]**

**26** A frustum is made by removing a smaller cone from the top of a larger cone.
The larger cone has base radius 10 cm and vertical height 24 cm.
The frustum has vertical height 15 cm.




**(a)** Find the radius of the top of the frustum.

..... cm **[2]**

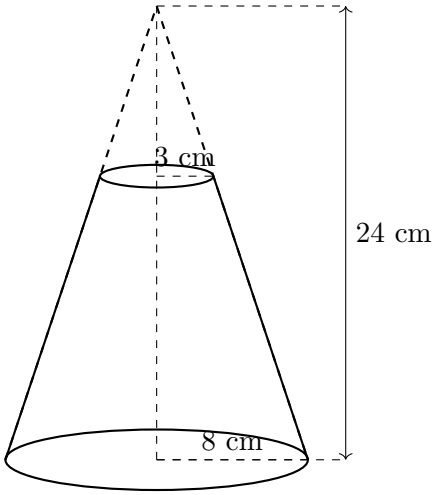
**(b)** Calculate the volume of the frustum.

..... $cm^3$ **[3]**

**(c)** Calculate the total surface area of the frustum.

..... $cm^2$ **[5]**

**27** A frustum is made by removing a small cone from the top of a large cone.
The large cone has a base radius of 8 cm and a total height of 24 cm.
The removed small cone has a base radius of 3 cm.




**(a)** Show that the height of the small cone removed is 9 cm.

**[2]**

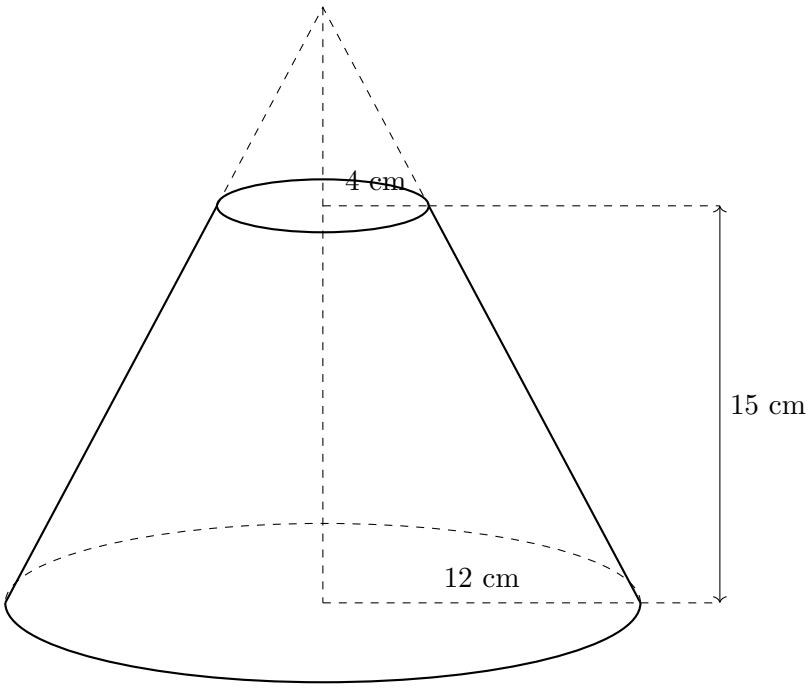
**(b)** Calculate the volume of the frustum.

Volume = ..... $cm^3$ **[3]**

**(c)** Calculate the total surface area of the frustum.

Surface area = ..... $cm^2$ **[5]**

**28** The diagram shows a solid frustum of a cone.
The frustum is made by removing a small cone from the top of a larger solid cone.
The base radius of the frustum is 12 cm and the top radius is 4 cm.
The height of the frustum is 15 cm.




**(a)** Calculate the height of the small cone that was removed.

Height = ..... cm **[3]**

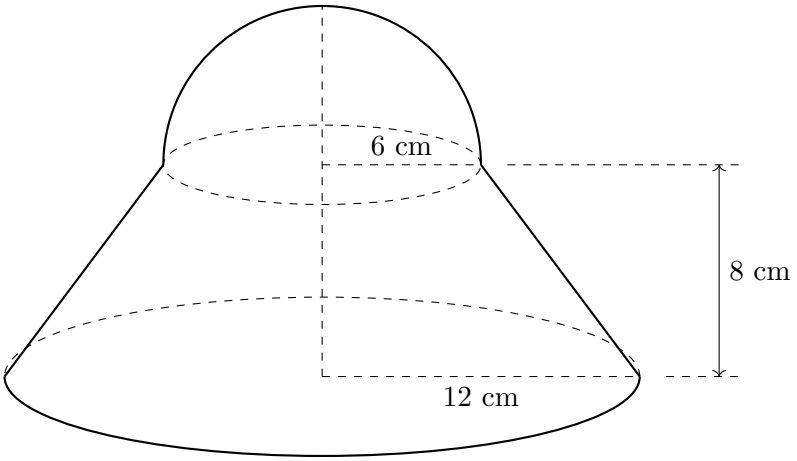
**(b)** Calculate the volume of the solid frustum.

Volume = ..... $cm^3$ **[3]**

**(c)** Calculate the total surface area of the solid frustum.

Surface area = ..... $cm^2$ **[5]**

**29** The diagram shows a compound solid made from a frustum of a cone and a solid hemisphere.
The frustum is created by removing a small cone from a large cone.
The base radius of the frustum is 12 cm and its top radius is 6 cm.
The height of the frustum is 8 cm.
A solid hemisphere of radius 6 cm is attached to the top face of the frustum.




NOT TO SCALE

**(a)** Show that the perpendicular height of the large cone from which the frustum is made is 16 cm.

**[3]**

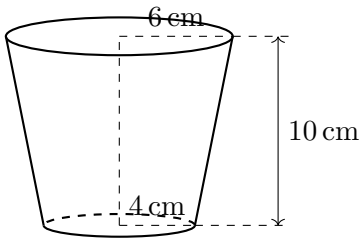
**(b)** Calculate the total volume of the compound solid.

..... $cm^3$ **[4]**

**(c)** Calculate the total surface area of the compound solid.

..... $cm^2$ **[4]**

**30** The diagram shows a closed container in the shape of a frustum of a cone.
The radius of the top circular face is 6 cm and the radius of the base is 4 cm.
The vertical height of the container is 10 cm.




NOT TO SCALE

**(a)** The frustum is made by removing a smaller cone from a larger cone.
Calculate the vertical height of the larger cone.

..... cm **[3]**

**(b)** Calculate the volume of the container.
Give your answer correct to 3 significant figures.

..... $cm^3$ **[3]**

**(c)** Calculate the total surface area of the closed container.
Give your answer correct to 3 significant figures.

..... $cm^2$ **[5]**